TriAlign

Progress in comparing clinical trial eligibility criteria

Jonathan Ma Jiatong Gao

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Project objective

Help a clinical team compare a current eligibility rule with rules from other trials.

1. Retrieve potentially comparable criteria.
2. Explain operational differences using source evidence.
3. Record a relationship or escalate an unresolved comparison.

Illustrative case: inclusion age ≥ 18 versus inclusion age < 18.
Shared words conceal disjoint allowed populations.

Scope: criterion comparison for protocol review, without patient screening.

Source: technical guide, Sections 1 and 3. Example is illustrative, not a measured encoder failure.

Current dataset

Source	Current artifact
Modern AutoCrit	Reviewed atomic criteria derived from public trial eligibility text
Frozen release	`trial100_v1`: 3,537 criteria across 100 trials
Review state	All retained rows marked corrected; zero recorded validation issues or readiness blockers
Provenance	Trial and criterion IDs, source versions, parent statements and available context
Reproducibility	Parquet snapshot with corpus hash and adjacent manifest

Criterion validation is complete by the source review record. **Clinical pair labels are a separate task.**

Source: trial100_v1.parquet.manifest.json. Review authority: project owner's reported manual validation. Chia remains an optional external source, not part of this release.

Pair construction and manual review

Completed: reproducible development pilot and an Excel labeling workbook.

Primary sampling method	Pairs
Same entity with structured differences	25
Further same-entity matches	25
Lexical neighbors within a domain	15
Random controls	10
Total natural cross-trial pairs	75

Twenty trials reserved for development. Family audit remains pending.

No pair labels entered in the workbook as of November 6.

Source: pair_pilot_v1/manifest.json, generator and labeling workbook. Sampling categories do not imply labels.

Clinical relationship labels

Comparable allowed populations

- Equivalent
- A more restrictive
- B more restrictive
- Partial overlap
- Contradictory or disjoint

Other dispositions

- Related, not comparable
- Unrelated
- Ambiguous

Interpret eligibility role, internal negation, units, time and Boolean scope before assigning a label.

Example: inclusion age ≥ 21 is more restrictive than inclusion age ≥ 18 in a common age frame.

Source: pair labeling contract 0.1 and technical guide, Section 3. Reversing A/B swaps only the two restrictive labels.

AI and agent workflow

Stage	Intended behavior	Current status
Candidate retrieval	Find cross-trial lexical and dense neighbors	Teammate branch
Exact tools	Compare supported, fully interpreted constraints	Starter comparator
Encoder and head	Rank candidates and predict ordered relationships	Fine-tuning and head planned
LLM judge and critic	Resolve contextual cases and challenge evidence	Planned
Orchestrator	Reconcile results, trace routes and escalate	Exact-only skeleton
Human review	Inspect evidence and record final decisions	Excel pilot available

Source: current orchestrator, structured tool and technical guide; teammate branch d30c71e for dense retrieval. Full routed workflow is not implemented.

Baseline progress

Code present on the teammate branch

- TF–IDF, MPNet and BioLORD retrieval interfaces.
- Shared neighbor format with query/candidate IDs, rank and score.
- Cross-trial filtering and normalized dense embeddings.

Teammate-reported integration results

- Common provisional sample: 200 criteria across 100 trials.
- 20 tests passed; saved-output checks passed.

Clinical performance is not yet measured. Validated relevance labels and exact resolved model revisions are still needed.

Source: teammate's progress report and local branch feature/comparisons-agent-ui at d30c71e. Saved run artifacts are absent from this checkout.

A failure in the current exact comparator

Constructed diagnostic	Rule
A	Inclusion: age $\geq$ 18 years
B	Exclusion: age $\geq$ 18 years
Current tool output	Equivalent, confidence 1.0
Expected relationship	Disjoint allowed age sets in this simplified frame

The comparator matches intervals without first applying eligibility role.
Its fixed confidence values do not measure calibrated certainty.

Next fix: gate context and polarity before allowing an exact label.

Source: constructed diagnostic executed against main on November 6; structured.py. This is a code failure case, not a corpus error rate.

Current challenges and limitations

- Pair labels and an audited family split are still missing.
- The starter exact tool does not implement the complete labeling contract.
- Baseline integration needs shared, versioned inputs and resolved model commits.
- We have no fine-tuned encoder, calibrated uncertainty or clinical evaluation yet.
- The UI, durable review workflow and end-to-end latency still need validation.

Local execution avoids website hosting. Optional external LLM calls still require network access and may incur usage costs.

Source: technical guide, local implementation and baseline branch review. Family/duplicate handling precedes held-out evaluation.

Next steps

Owner	Deliverable before the final presentation
Jonathan	Adjudicated pilot, revised guide and audited family map
Jiatong	Baseline PR and reproducible runs with pinned revisions
Together	Local pair-review UI with persistent save/reload
Together	Baseline and exact-tool metrics with documented errors
Jiatong	Fine-tuning and held-out comparison, if labels and development results justify training

Final presentation target: December 8, subject to course confirmation.
Training remains conditional on label quality and development evidence.

Source: technical guide milestones and pair labeling contract. Dates are planning targets, not completed milestones.

Questions and feedback

Which clinical comparisons should the pilot prioritize?

What evidence would make a comparison useful to a protocol reviewer?

What should a minimum useful demonstration show?